Implementation Documentation

Write a short 1-2 page explanation which:

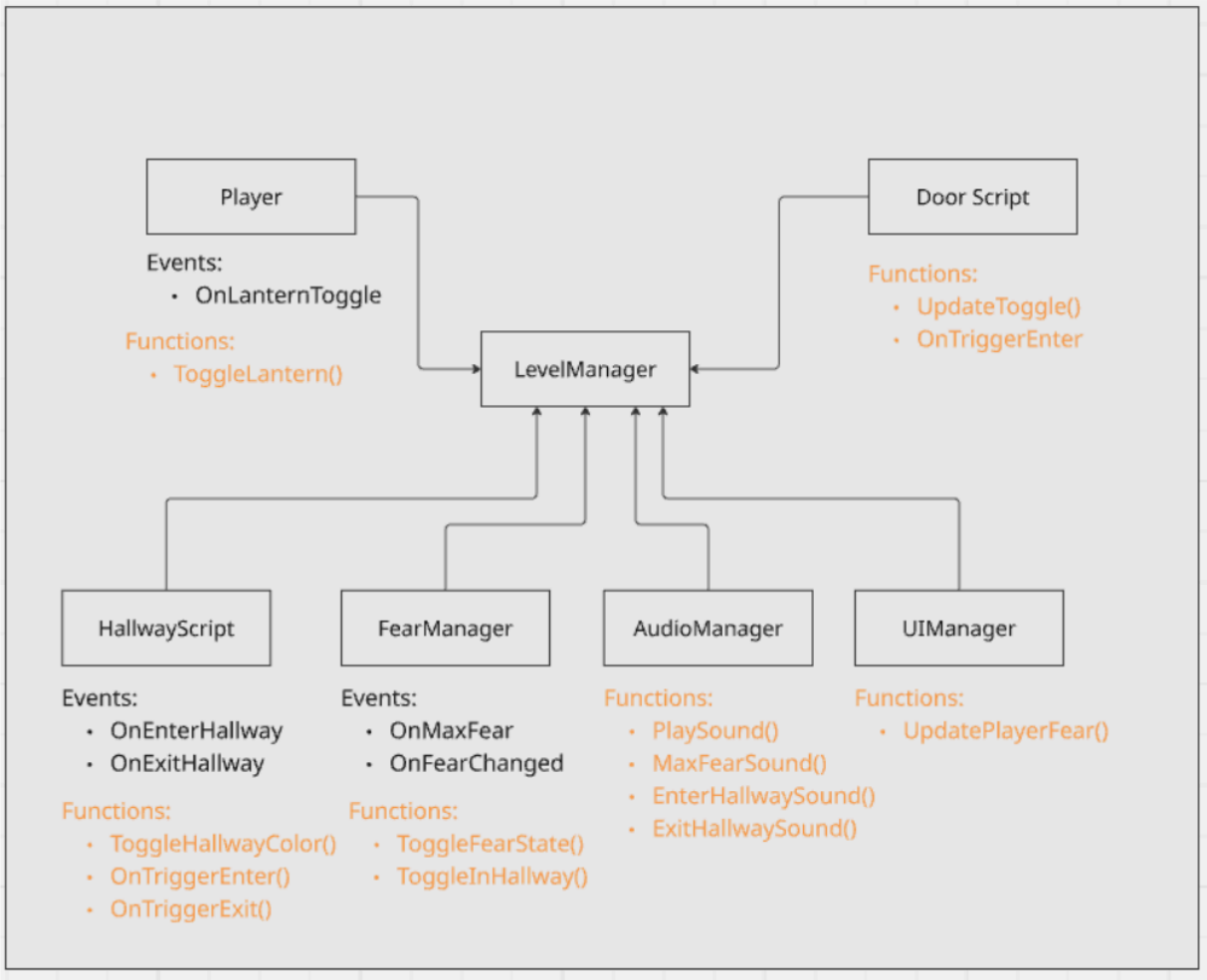
- *Describes the event flow of the system.*
- *Explains how events improve modularity and gameplay responsiveness.*
- *Reflects on challenges and solutions during development.*

Assignment #2: Event-Driven Systems Implementation

GAME10020

Becca Pesner

System Layout Diagram



System Event Flow

The system communicates through the MyLevelManager script, which adds all event listeners in the Start() function. The major unity event is the OnLanternToggled event on the Character script, allowing the wall barrier to toggle position, the hallway to darken and lighten, as well as controlling the lock on the level exit door. Additional components such as the audio manager simply trigger a PlaySound function when a given event invocation appears.

Fear Manager Listeners:

- LevelManager.cs - End Level Function
    - Listens to the OnMaxFear event
- AudioManager.cs - Max Fear Sound Function
    - Listens to the OnMaxFear event
- UIManager.cs - Update Player Fear Function
    - Listens ot the OnFearChanged event

Player Listeners:

- FearManager.cs - Toggle Fear State Function
    - Listens to the OnLanternToggle event
- DoorScript.cs - Update Toggle Function
    - Listens to the OnLaternToggle event
- HallwayScript.cs - Toggle Hallway Colour Function
    - Listens to the OnLanternToggle event

Hallway Script Listeners:

- FearManager.cs - Toggle in Hallway Function
    - Listens to the OnExitHallway event
- AudioManager.cs - Exit Hallway Sound Function
    - Listens to the OnEnterHallway event

Modularity and Gameplay Responsiveness

Using Unity Events has allowed me to create systems that are modular and decoupled, allowing me to add or remove components without the entire system breaking. This allows for quick and easy prototyping of additional components within the system, as I

am able to enable and disable components without getting null reference exception errors. As for gameplay responsiveness, reducing the number of function calls during the update loop allows for less memory usage per frame. This produced a smoother gameplay result, however, the limited amount of objects within this Unity Scene does not have much to affect performance in the first place. The use of these modular systems will have a significantly larger impact on performance with a larger project.

Challenges and Solutions

I have designed the system to communicate all events through the MyLevelManager script. Every script has been created in addition to the module example, with the exception of the Character.cs script, in which I have modified to create an Unity Event “OnLanternToggle”. I could have created an additional player script to handle the OnLanternToggle event, but I decided to use the already present “attack” bool found within the Character script. Other than understanding how the event system is, I found it rather easy to create the functionality that I have implemented.